

EBAU EZ OHIKOA 2024

B3 Ariketa

Izan bedi $f(x) = 2xe^{-2x^2}$. $\Rightarrow \text{Dom } f = \mathbb{R}$.

(a) (1 p) Aurkitu f -ren gorakortasun- eta beherakortasun-tarteak.

(b) (1 p) Aurkitu f -ren mutur erlatiboak, eta arrazoitu maximoak edo minimoak diren.

(c) (0,5 p) Aurkitu f -ren asintotak.

a) GORAKORTASUN ETA BEHERAKORTASUNA

$$f'(x) = 2 \cdot e^{-2x^2} + 2x \cdot (-4x) \cdot e^{-2x^2} = e^{-2x^2} (2 - 8x^2)$$

$$f'(x) = 0 \Rightarrow e^{-2x^2} (2 - 8x^2) = 0$$

$$\begin{cases} e^{-2x^2} \neq 0 \\ 2 - 8x^2 = 0 \rightarrow x = \pm 1/2 \end{cases}$$

$$\begin{array}{c} f'(x) \quad f' < 0 \quad | \quad f' > 0 \quad | \quad f' < 0 \\ f(x) \quad \searrow \quad \nearrow \quad \searrow \end{array}$$

$$\text{G.T. } (-1/2, 1/2)$$

$$\text{B.T. } (-\infty, -1/2) \cup (1/2, +\infty)$$

b) MUTUR ERLATIBOAK

Mutur erlatiboak $f'(x) = 0$ $\begin{cases} \text{Maxim. } f''(x_0) < 0 \\ \text{Minim. } f''(x_0) > 0 \end{cases}$

$$f''(x) = -4x e^{-2x^2} (2 - 8x^2) + e^{-2x^2} (-16x)$$

$$f''(x) = e^{-2x^2} (32x^3 - 24x)$$

$$f''(1/2) = e^{-2 \cdot 1/4} (32 \cdot (1/2)^3 - 24 \cdot (1/2)) = -8/\sqrt{e} < 0 \quad \underline{\text{MAX}}$$

$$f''(-1/2) = e^{-2 \cdot 1/4} (32 \cdot (-1/2)^3 - 24 \cdot (-1/2)) = 8/\sqrt{e} > 0 \quad \underline{\text{MIN}}$$

$$f(1/2) = 2 \cdot 1/2 \cdot e^{-2 \cdot (1/2)^2} = e^{-1/2}$$

$$f(-1/2) = 2 \cdot (-1/2) \cdot e^{-2 \cdot (-1/2)^2} = -1/\sqrt{e}$$

Maxim. erlatib. $(1/2, e^{-1/2})$

Minim. erlatib. $(-1/2, -1/\sqrt{e})$

c) ASINTOTAK.

$$\text{Dom } f = \mathbb{R}.$$

ASINT. BERTKALAK $\lim_{x \rightarrow a} f(x) = \pm \infty \Rightarrow \nexists$ asintoten
bertkalk

ASINT. HORIZONTAL/ZEHORRIG...

$$\lim_{x \rightarrow +\infty} 2x \cdot e^{-2x^2} = \lim_{x \rightarrow +\infty} \frac{2x}{e^{2x^2}} = \frac{+\infty}{+\infty} = 0$$

Zufutok kourvatur, esponentialen
 ∞ -a walo jarekoo da.

ASINTOTA HORIZONTALA $y=0$.

$$\lim_{x \rightarrow -\infty} 2x \cdot e^{-2x^2} = \lim_{x \rightarrow +\infty} 2(-x) e^{-2(x)^2} =$$

$$= \lim_{x \rightarrow +\infty} -2x \cdot e^{-2x^2} = \lim_{x \rightarrow +\infty} \frac{-2x}{e^{-2x^2}} = \frac{-\infty}{+\infty} = 0.$$